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RECOVERY ARCHITECTURE

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Solving the Multiplier Effect

Agentic AI and discrete-event simulation engine for multi-day airline network recovery.

NETWORK CASCADE

The Multiplier Blind Spot

A local disruption cascades into massive network-wide financial losses.

- Direct cost of a 2-hour runway reduction at ORD is **$181K**.
- This disruption cascades into a severe **$942K** network loss.
- The result is a **5.19x multiplier** effect that traditional tools fail to mitigate.

5.19×

The Multiplier Effect

Traditional tools fail to mitigate this compounding network cascade.

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AGENTIC SIMULATION

Arbitrated Financial Simulation

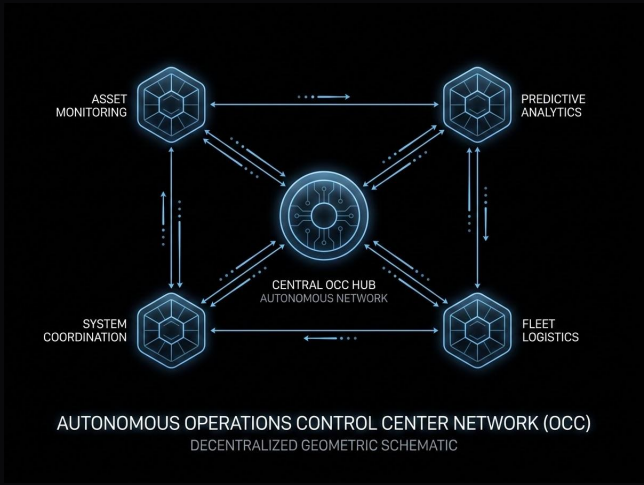
Agentic simulation outperforms traditional rule-engines in adversarial recovery.

- **5-Agent OCC:** Powered by a specialized 5-agent Operations Control Center.
- **Duty Manager:** Executes parallel financial simulations to optimize recovery paths.

ADVERSARIAL RECOVERY

$145,854

Proven savings over traditional rule-engines in adversarial recovery tests.



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SYSTEM ARCHITECTURE

Gemini Proposes. Code Disposes.

A strict operational boundary separating natural language reasoning from deterministic execution layers.

01 / PARSE

Gemini Proposes

Gemini parses natural language directly into strict Pydantic schemas.

02 / EXECUTE

Code Disposes

A deterministic code engine executes all mathematical and simulation logic.

03 / ARBITRATE

Zero Hallucination

This strict boundary ensures zero hallucination during critical operational arbitration.

Enterprise-Grade Verification

SkyRipple operates at massive scale with zero latency.

582,304

Grounded: Validated against 582,304 real BTS records (Dec 2025) and FAA registries.

$5.8M

Emergent Insights: Surfaced counter-intuitive network windfalls, including a $5.8M cost swing from competitor groundings at DEN.

8s

Memoization: Aggressive caching dropped massive multi-day passenger itinerary generation from 110s down to 8s.

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The $30B Efficiency Gap

Multi-day recovery represents a massive global opportunity.

Market Breakdown

- **TAM (Global):** $30B+ lost annually to systemic delays across global commercial aviation and logistics.
- **SAM (US Domestic):** Major carriers operating complex, high-vulnerability hub-and-spoke models.
- **SOM (Top 5 Airlines):** The largest US operators facing immediate FAA regulatory pressure and crew-legality bottlenecks.



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Deterministic Intelligence

Real-time financial simulation surpasses static rule-engines.

Legacy Rules

Mathematically reliable, but rely on blunt, static logic that breaks during multi-day cascades.

Standard LLMs

Highly adaptable and intelligent, but structurally prone to hallucinating financial outcomes.

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Occupies the only viable enterprise quadrant—adaptive intelligence through deterministic, safety-by-construction arbitration.

BUSINESS MODEL

Scalable licensing aligns with operational compute demands.

PLATFORM

Core OCC SaaS

Annual recurring seat & hub license for Airline Operations Control Centers.

Includes live disruption parsing, real-time cascade visualization, and standard single-day recovery.

COMPUTE & SCALE

Multi-Agent Simulation

Tiered/consumption billing for parallel continuous-month simulations and heavy agentic arbitration.

Dedicated high-throughput VPC instances for zero-latency operational recovery.

PERSPECTIVE

Engineering Meets Strategy

Bridging computer science with operational management.

- Built by **Aaryan Sharma**, Systems Engineer.
- Master in Management candidate at **Kellogg School of Management**.
- Focused on solving **multi-million-dollar** operational bottlenecks.



Aaryan Sharma

Systems Engineer

Master in Management
Kellogg School of Management

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The Next Era of Ops

Partner with us to orchestrate global operational recovery.

Seeking agreements with pilot airlines for initial live deployment.

Validating technical superiority in real-world multi-day recovery environments.

Establishing the definitive path to global operations orchestration.

Our Vision

"Orchestrating Global Operational Recovery"

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